

RAG Ingestion

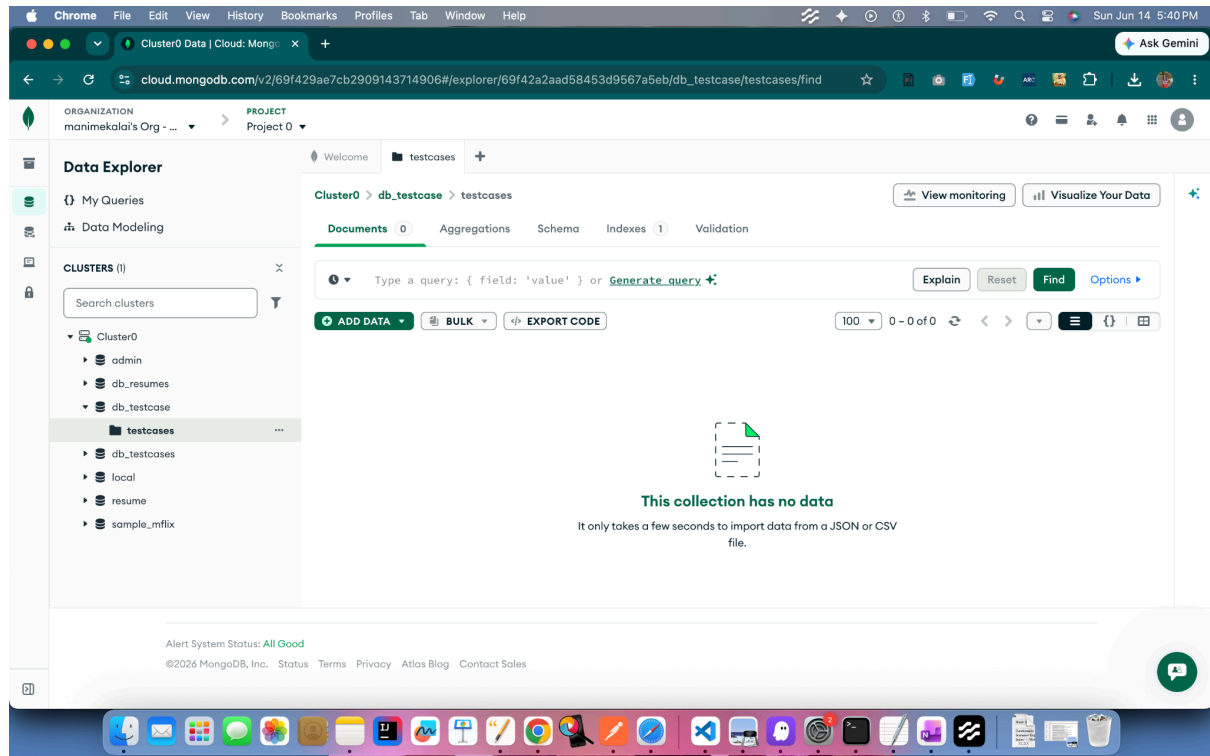
Step 1:

Before creating the DB and collection, cluster0 should be created

Create a database and a collection in MongoDB

DB name: db_testcase

Collection: testcases



Step 2: Creation of vector search

1. Select Search & Vector Search
2. Click on Create Search Index
3. Select vector search from index configuration
4. Select "Bring your own embeddings"
5. Index name as "vector_index_testcases"
6. Select database and collection db_testcase and testcases

Cluster0 Search | Cloud: Mon x + Ask Gemini

cloud.mongodb.com/v2/69f429ae7cb2909143714906#/clusters/atlasSearch/Cluster0

Search & Vector Search

Find a search index

VIEW COMPOUND QUERY EXAMPLE + CREATE SEARCH INDEX

M0 free clusters have a limit of 3 search and vector indexes per cluster. Indexes Used: 2 of 3.

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Est)
db_testcases	test_cases	bm25_indexsearch	READY View status details	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id" + 9 more	6,354 (100%) inc 6,354
resume	resume_collection	resume_vectorsearch	READY View status details	✓	vectorSearch	"embedding", "id", "name", "email", "phone", + 14 more	12 (100%) indexe

[Learn more about Atlas Search](#)
[Learn more about Atlas Vector Search](#)

Alert System Status: All Good
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Cluster0 Search | Cloud: Mon x + Ask Gemini

cloud.mongodb.com/v2/69f429ae7cb2909143714906#/clusters/atlasSearch/Cluster0?view=Create

Search & Vector Search

testcases

- db_testcases
- resume
- sample_mflix

Configuration Method

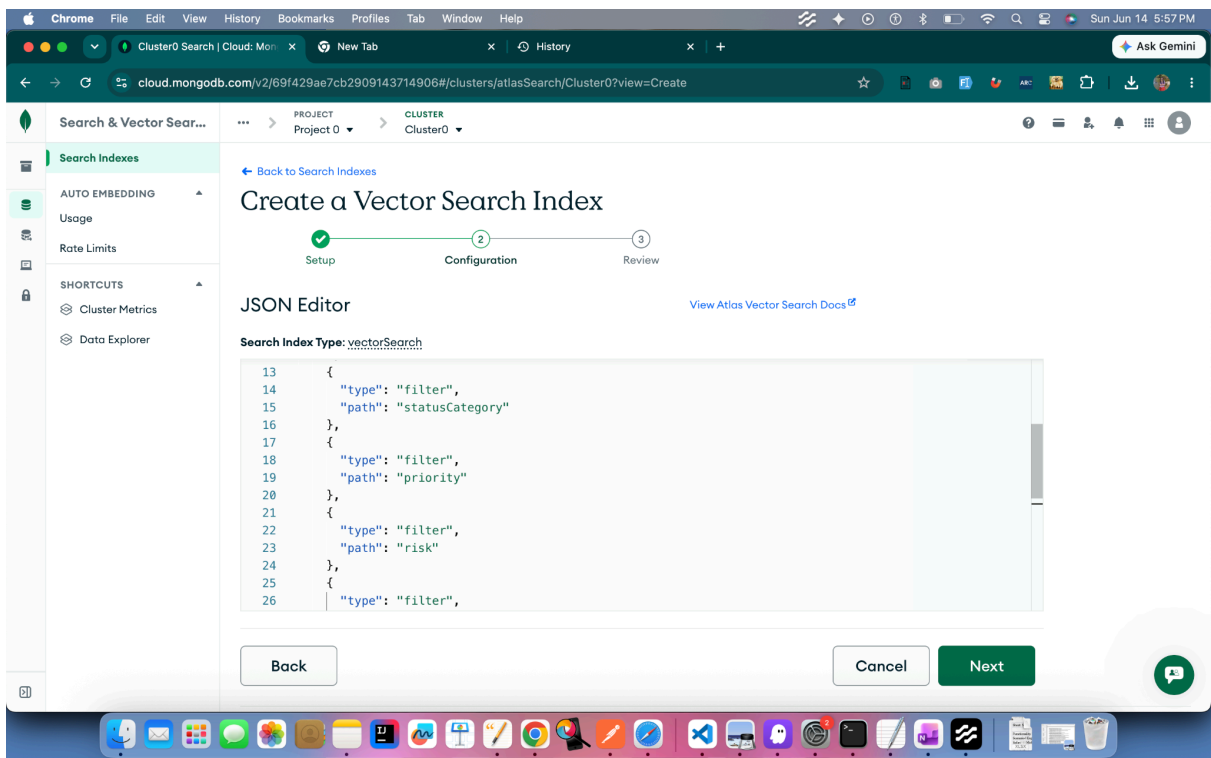
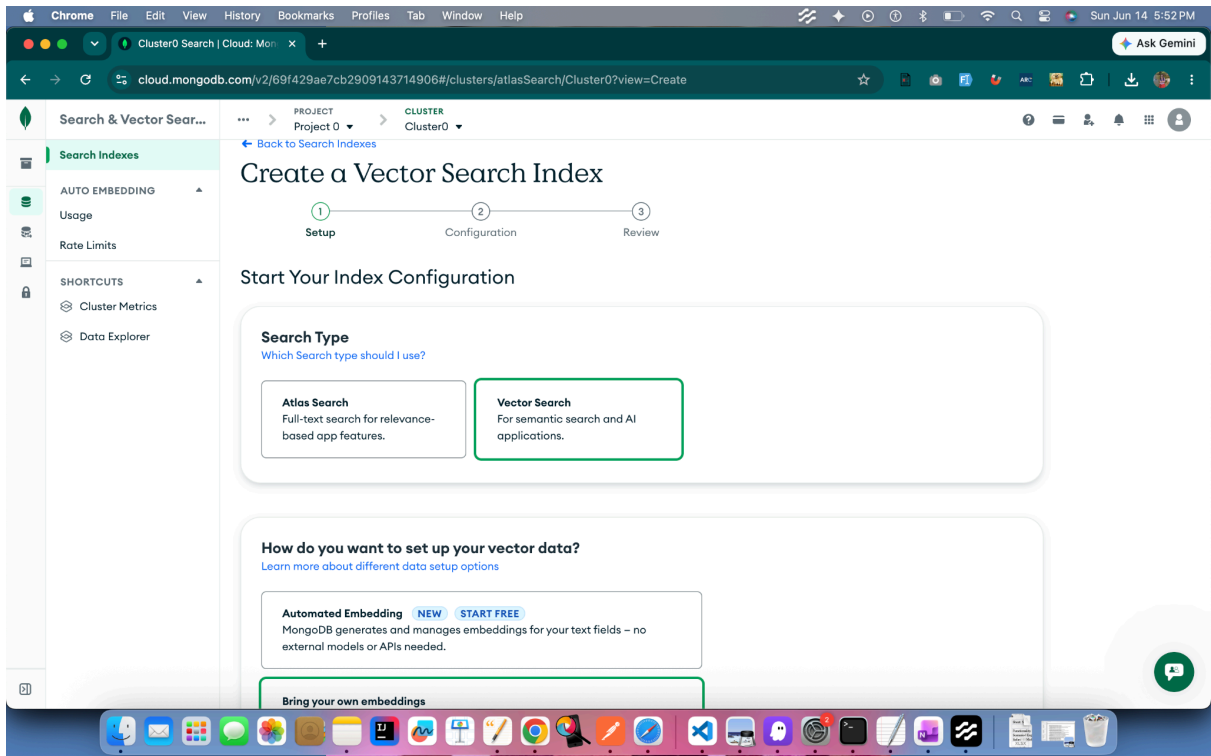
Select how you would like to configure your index. You can also define indexes through the [Atlas Admin API](#), [Atlas CLI](#), [MongoDB Shell](#), or supported [Drivers](#).

Visual Editor
Define index definitions in a more guided experience.

JSON Editor
Edit the raw index definition with an embedded JSON editor.

Cancel Next

Alert System Status: All Good
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Import the file

Open a terminal and npm install(dependencies will be installed)

.env file Update the values below

MONGODB_URI

DB_NAME

COLLECTION_NAME

VECTOR_INDEX_NAME

MISTRAL_API_KEY

The screenshot shows a VS Code editor interface with a file explorer on the left, a central editor window, a terminal at the bottom, and a chat window on the right.

File Explorer: Shows the project structure with folders like `client`, `node_modules`, `releases`, `server`, `src`, and `uploads`. The `.env` file is selected.

Editor Window: Displays the contents of the `.env` file, which includes configuration for MongoDB, Mistral AI, and Groq AI.

```
1 MONGODB_URI=mongodb+srv://manimekalai:manimekalai190495@cluster0.cg5tty.mongodb.net/?retryWrites=true&ssl=true&authSource=admin
2 #MONGODB_URI=mongodb+srv://sumitha:TestLeaf321@cluster0.c5l4wj.mongodb.net/?retryWrites=true&ssl=true&authSource=admin
3 DB_NAME="db_testcase"
4 #DB_NAME="RagTest_cases"
5
6 COLLECTION_NAME="testcases"
7 VECTOR_INDEX_NAME="vector_index_testcases"
8 BM25_INDEX_NAME="bm25_search"
9
10 USER_STORIES_COLLECTION_NAME="user_stories"
11 USER_STORIES_VECTOR_INDEX_NAME="vector_index_user_story"
12
13 #USER_STORIES_COLLECTION_NAME="user_stories"
14 #USER_STORIES_VECTOR_INDEX_NAME="vector_index"
15
16 # Mistral AI Configuration (for embeddings)
17 MISTRAL_API_KEY=pjXDT57jIAEd7rPTqHmQ3XYXhTLLfcj
18 MISTRAL_EMBEDDING_MODEL="mistral-embed"
19
20 # Groq AI Configuration (for reranking and summarization)
21 GROQ_API_KEY=gsK_102wSghJJRZ7b2UgmePtWgdyb3FY0gy0o06hEZ0yLBbElz8n5t7
22 GROQ_RERANK_MODEL="openai/gpt-oss-120b"
23 GROQ_SUMMARIZATION_MODEL="openai/gpt-oss-120b"
```

Terminal: Shows the output of a batch script, including token counts and cost savings.

```
[0] Total Cost: $0.009581
[0] [testcases] Total Tokens: 95,809
[0] Average Cost per Test Case: $0.00001916
[0] [testcases] Average Tokens per Test Case: 192
[0] Cost Savings: ~90% cheaper than OpenAI embeddings
[0] Batch script completed successfully!
```

Chat Window: Contains instructions and terminal commands for running the application.

Instructions:

- I'll restart the dev servers using a different client port (PORT=3002) to avoid killing any running process; expected outcome: server on 3001 and client on 3002 both running. I'll run the command now.
- I'll (1) check which processes are listening on ports 3001 and 3002, (2) stop any existing server process, then (3) start dev with the server on 3001 and the client on 3002 so they don't conflict; expected outcome: server on 3001 and client on 3002 both running. I'll run these steps now.
- I'll check which processes are listening on ports 3000-3002, kill any node/react processes found there, then restart `npm run dev` so server and client can start cleanly. I'll run those commands now and report the outcome.

Commands:

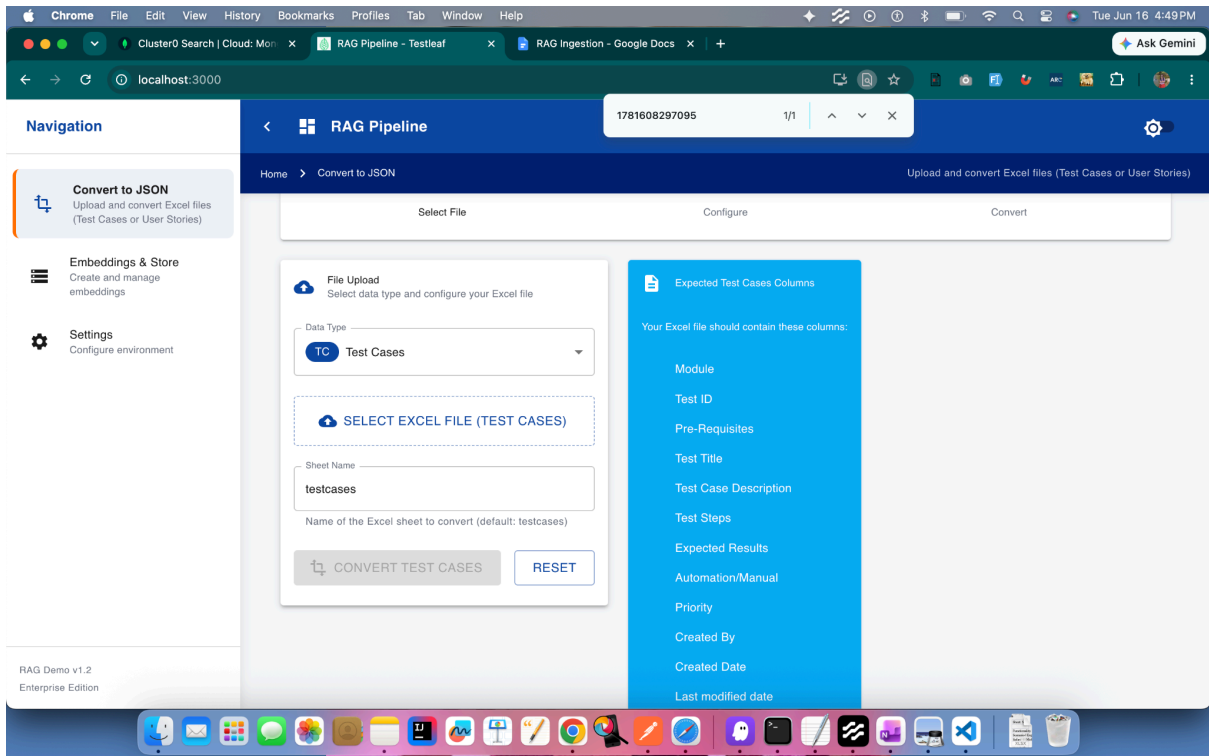
```
cd "/Users/meghavela/Desktop/Weekly Topics/week3/rag-mongo-demo-ui" && PORT=3002 npm run dev

cd "/Users/meghavela/Desktop/Weekly Topics/week3/rag-mongo-demo-ui" && lsof -nP -iTCP:3001 -sTCP:LISTEN || true && lsof -nP -iTCP:3002 -sTCP:LISTEN || true && pkill -f "node server/index.js" || true && sleep 1 && concurrently "PORT=3001 npm run server" "PORT=3002 npm run client"
```

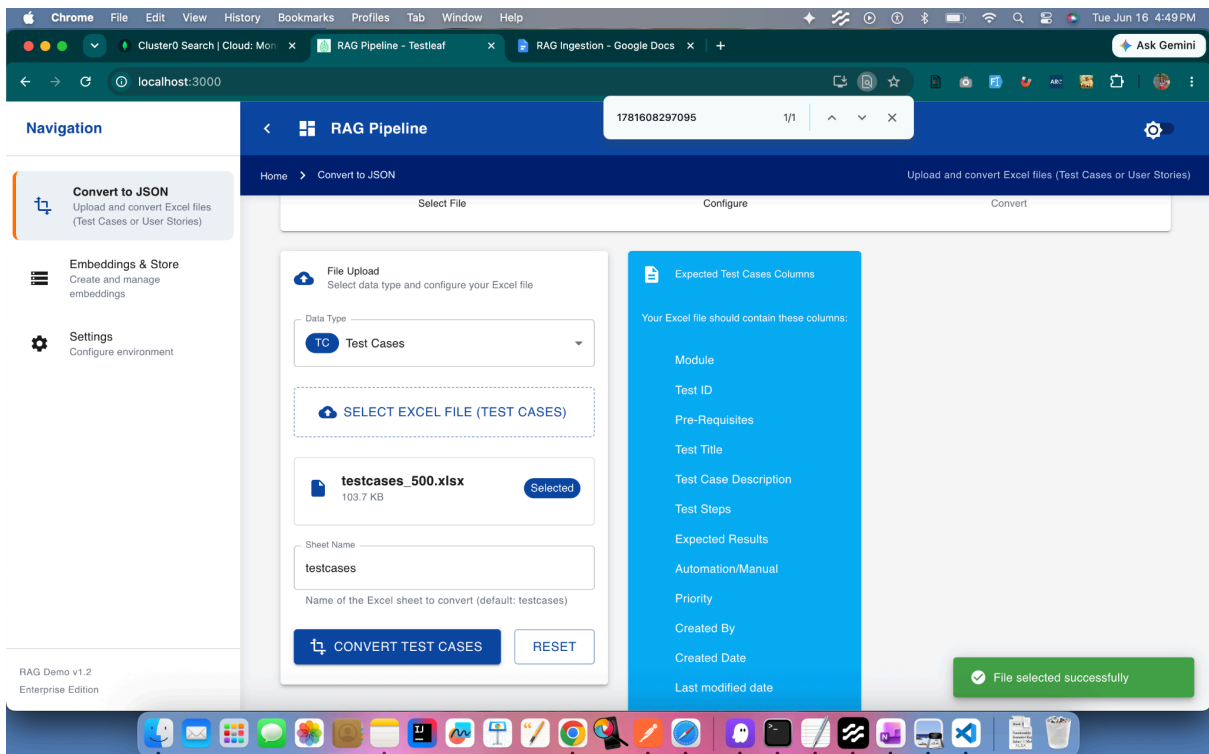
```
cd "/Users/meghavela/Desktop/Weekly Topics/week3/rag-mongo-demo-ui" && echo 'Listening before:' && lsof -nP -iTCP:3000,3001,3002 -sTCP:LISTEN || true && PIDS=$(lsof -nP -iTCP:3000,3001,3002 -sTCP:LISTEN 2>/dev/null | awk 'NR>1{print $2}' | sort -u) && if [ -n "$PIDS" ]; then echo "Killing PIDs: $PIDS" && kill $PIDS || kill -9 $PIDS; else echo "No PIDs found to kill"; fi && sleep 1 && echo 'Listening after kill!' && lsof -nP -iTCP:3000,3001,3002 -sTCP:LISTEN || true && npm run dev
```

Start npm run dev

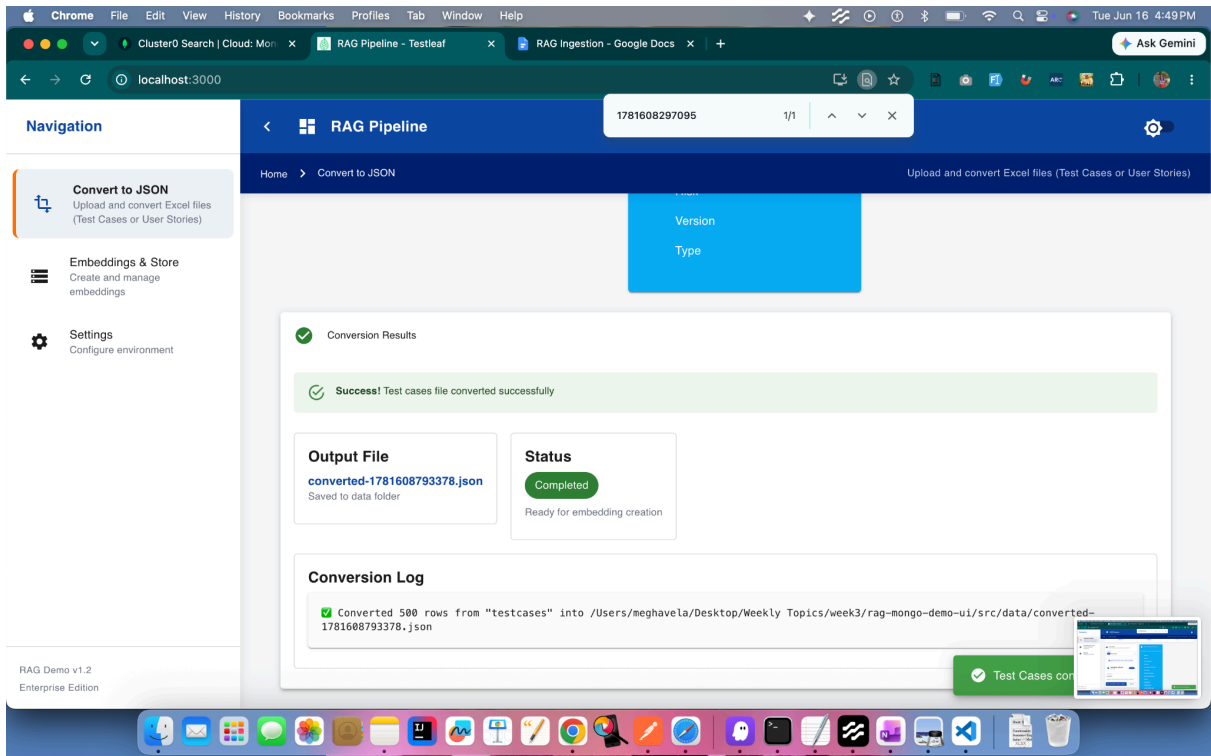
The server will be started



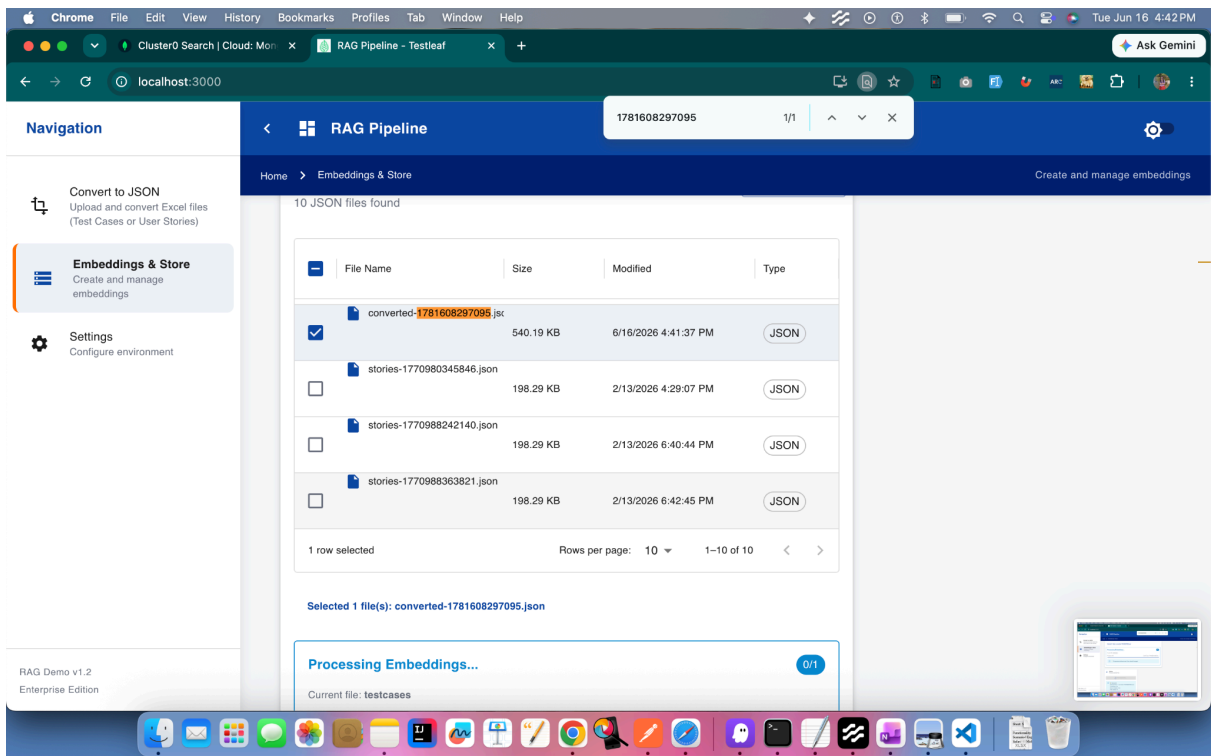
Select and upload the testcases and click convert to JSON



File will be converted to json



After converting select Embeddings & store
Select the converted json to Embeddings



Check file is getting updated in the DB

The screenshot shows a VS Code editor with a terminal window displaying test results for 'testcases-vector-index.json'. The test results show that 500 testcases are being processed in batches, with success rates and costs reported. A chat window on the right provides instructions for restarting the dev servers using a different client port (PORT=3002) to avoid conflicts.

Terminal Output:

```
[0] [testcases] [Batch 1/10] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 2/10] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 3/10] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 2/10] Success! Cost: $0.000965 | Tokens: 9649
[0] [testcases] [Batch 1/10] Success! Cost: $0.000768 | Tokens: 7677
[0] [testcases] [Batch 3/10] Success! Cost: $0.000942 | Tokens: 9420
[0] [testcases] Waiting 1000ms before next batch group...
[0] [testcases] [Batch 4/10] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 6/10] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 4/10] Success! Cost: $0.000980 | Tokens: 9797
[0] [testcases] [Batch 6/10] Success! Cost: $0.001028 | Tokens: 10279
[0] [testcases] [Batch 5/10] Success! Cost: $0.001194 | Tokens: 11938
[0] [testcases] Waiting 1000ms before next batch group...
[0] [testcases] [Batch 7/10] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 8/10] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 9/10] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 7/10] Success! Cost: $0.001092 | Tokens: 10918
```

Chat Instructions:

I'll restart the dev servers using a different client port (PORT=3002) to avoid killing any running process; expected outcome: server on 3001 and client on 3002 both running. I'll run the command now.

```
cd "/Users/meghavela/Desktop/Weekly Topics/week3/rag-mongo-demo-ui" && PORT=3002 npm run dev
```

I'll (1) check which processes are listening on ports 3001 and 3002, (2) stop any existing server process, then (3) start dev with the server on 3001 and the client on 3002 so they don't conflict; expected outcome: server on 3001 and client on 3002 both running. I'll run these steps now.

```
cd "/Users/meghavela/Desktop/Weekly Topics/week3/rag-mongo-demo-ui" && lsof -nP -iTCP:3001 -sTCP:LISTEN || true && lsof -nP -iTCP:3002 -sTCP:LISTEN || true && pkill -f "node server/index.js" || true && sleep 1 && concurrently "PORT=3001 npm run server" "PORT=3002 npm run client"
```

I'll check which processes are listening on ports 3000-3002, kill any node/react processes found there, then restart npm run dev so server and client can start cleanly. I'll run those commands now and report the outcome.

```
cd "/Users/meghavela/Desktop/Weekly Topics/week3/rag-mongo-demo-ui" && echo 'Listening before:' && lsof -nP -iTCP:3000,3001,3002 -sTCP:LISTEN || true && PIDS=$(lsof -nP -iTCP:3000,3001,3002 -sTCP:LISTEN 2>/dev/null | awk 'NR>1{print $2}' | sort -u) && if [ -n "$PIDS" ]; then echo 'Killing PIDS: $PIDS' && kill $PIDS || kill -9 $PIDS; else echo 'No PIDS found to kill'; fi && sleep 1 && echo 'Listening after kill:' && lsof -nP -iTCP:3000,3001,3002 -sTCP:LISTEN || true && npm run dev
```

Check 500 testcases are getting updated in the DB

The screenshot shows the MongoDB Atlas Search & Vector Search interface. It displays a table of search indexes with columns for Status, Queryable, Type, Index Fields, Documents (Estimated), Size, Required Memory, and Actions. The table shows three indexes: a vectorSearch index for 'embedding', a search index for 'automationManual', and a vectorSearch index for 'embedding', 'id', 'name', 'email', and 'phone'.

Status	Queryable	Type	Index Fields	Documents (Estimated)	Size	Required Memory	Actions
READY	✓	vectorSearch	"embedding", "projectName", "statusCategory", "priority", "risk".*2 more	500 (100%) indexed of 500	2.03MB	2.03MB	QUERY ...
READY	✓	search	"automationManual", "createdBy", "description", "expectedResults", "id".*2 more	6,354 (100%) indexed of 6,354	3.77MB	—	QUERY ...
READY	✓	vectorSearch	"embedding", "id", "name", "email", "phone".*14 more	12 (100%) indexed of 12	46.99KB	46.99KB	QUERY ...